Notebook 5 — Chapter 5.1

Joint, Marginal & Conditional Probability Distributions

20 worked examples — multivariate densities, the independence criterion, conditioning — calibrated for Stage 1/Stage 2 speed and rigor

A. Joint Distributions & Marginals

Example 1 — Reading Marginals Off a Discrete Table

A joint PMF table for (X, Y), $X \in \{1, 2\}$, $Y \in \{1, 2, 3\}$:

	$Y{=}1$	$Y{=}2$	$Y{=}3$
$X{=}1$	0.10	0.15	0.05
$X{=}2$	0.20	0.30	0.20

Find $f_X(x)$ and $f_Y(y)$.
Solution. Sum each *row* for f_X, each *column* for f_Y:

$$f_X(1) = 0.10 + 0.15 + 0.05 = 0.30, \quad f_X(2) = 0.20 + 0.30 + 0.20 = 0.70$$

$$f_Y(1) = 0.30,\ f_Y(2) = 0.45,\ f_Y(3) = 0.25$$

Check: both marginals sum to $\boxed{1}$.
⋆ *Analogy:* the joint table is a restaurant's order log, rows = main course, columns = side dish. "How popular is steak, regardless of side?" is exactly $f_X(\text{steak})$ — sum across the row. Marginalizing is just collapsing a spreadsheet pivot table along one axis.

Example 2 — A Non-Separable Continuous Density

$f_{X,Y}(x, y) = x + y$ on the unit square $[0, 1] \times [0, 1]$. Verify this is a valid density and find $f_X(x)$.
Solution. Normalization: $\int_0^1 \int_0^1 (x+y)\, dy\, dx = \int_0^1 \left(x + \frac{1}{2}\right) dx = \frac{1}{2} + \frac{1}{2} = \boxed{1}$. Marginal (integrate out y):

$$f_X(x) = \int_0^1 (x + y)\, dy = x + \frac{1}{2}, \quad 0 \le x \le 1$$

By symmetry $f_Y(y) = y + \frac{1}{2}$.
▶ *Analogy:* think of $f(x, y)$ as a tilted table top sitting over the unit square — higher in the far corner $(1, 1)$, lower near $(0, 0)$. Integrating out y is literally *slicing* the table top at a fixed x and measuring the *area* of that slice — a thin vertical strip's total "height budget."

Example 3 — Marginal With an x-Dependent Limit

$f_{X,Y}(x, y) = 2$ on the triangle $\{x > 0,\ y > 0,\ x + y < 1\}$ (uniform over the triangle). Find $f_X(x)$.
Solution. For fixed $x \in (0, 1)$, y ranges over $(0, 1 - x)$:

$$f_X(x) = \int_0^{1-x} 2\, dy = \boxed{2 - 2x}, \quad 0 < x < 1$$

Check: $\int_0^1 (2 - 2x) dx = 2 - 1 = 1$ ✓.
⋆ *Analogy:* slicing a triangular pizza vertically at position x — the slice gets *shorter* as x grows, because the triangle narrows. The marginal density inherits that narrowing: $f_X(x)$ falls linearly from 2 at $x = 0$ to 0 at $x = 1$. When the support itself isn't a rectangle, the integration limit depends on x — the single most common bug source in this chapter.

Example 4 — Solving for the Normalizing Constant

$f_{X,Y}(x, y) = c(x^2 + y)$ on $[0, 1] \times [0, 1]$. Find c.
Solution.

$$\int_0^1 \int_0^1 c(x^2 + y)\, dy\, dx = c \int_0^1 \left(x^2 + \tfrac{1}{2}\right) dx = c\left(\tfrac{1}{3} + \tfrac{1}{2}\right) = c \cdot \tfrac{5}{6}$$

Set equal to 1:

$$c = \boxed{\tfrac{6}{5}}$$

▶ *Speed trick: always* find c by double-integrating the *un-normalized* shape over its full support and setting the result to 1 — never guess $c = 1$ just because the formula "looks like" a density. This is the continuous twin of checking a discrete table sums to 1.

Example 5 — Recovering a Missing Cell, Then Marginals

A joint PMF table for two correlated order-flow indicators $X, Y \in \{0, 1\}$ has one entry unknown:

	$Y{=}0$	$Y{=}1$
$X{=}0$	0.25	0.15
$X{=}1$	0.30	p

Find p, then f_X and f_Y.
Solution. All four cells must sum to 1: $0.25 + 0.15 + 0.30 + p = 1 \Rightarrow p = \boxed{0.30}$.

$$f_X(0) = 0.40,\ f_X(1) = 0.60, \qquad f_Y(0) = 0.55,\ f_Y(1) = 0.45$$

⋆ *Analogy:* a 2×2 joint table has 4 cells but only 3 *independent* numbers — exactly like a weather forecast that must say "rain" or "no rain" with probabilities summing to 1. The missing cell is never "extra" information; it's forced by the other three, the same way the last row of any complete probability table is forced by the rest.

B. The Independence Criterion

Example 6 — Confirming Independence on a Table

$X, Y \in \{1, 2\}$ with joint PMF $f(1,1) = 0.12$, $f(1,2) = 0.28$, $f(2,1) = 0.18$, $f(2,2) = 0.42$. Are X, Y independent?
Solution. Marginals: $f_X(1) = 0.40$, $f_X(2) = 0.60$; $f_Y(1) = 0.30$, $f_Y(2) = 0.70$. Check *every* cell:

$$f_X(1) f_Y(1) = 0.12\checkmark, \quad f_X(1) f_Y(2) = 0.28\checkmark$$

$$f_X(2) f_Y(1) = 0.18\checkmark, \quad f_X(2) f_Y(2) = 0.42\checkmark$$

All four match $\Rightarrow$ $\boxed{\text{independent}}$.
▶ *Speed trick:* for a small table, independence holds iff *every row is a scalar multiple of every other row* (here row 2 = 1.5× row 1). Spotting that proportionality is faster than multiplying out all four marginal products.

Example 7 — The One-Cell Trap

Same table shape, $f(1,1) = 0.12$, $f(1,2) = 0.28$, $f(2,1) = 0.18$, $f(2,2) = 0.40$ (only the last cell changed, and rows/columns still sum sensibly: row sums $0.40, 0.58$, column sums $0.30, 0.68$, total 0.98 — rescale isn't the point here). Using the *un-normalized* relative weights, check whether X, Y are independent.
Solution. $f_X(1) = 0.40$, $f_X(2) = 0.58$; $f_Y(1) = 0.30$, $f_Y(2) = 0.68$ (as given weights). Test the changed cell:

$$f_X(2) f_Y(2) = 0.58 \times 0.68 = 0.3944 \neq 0.40 = f(2,2)$$

One mismatch is enough: $\boxed{\text{dependent}}$ — even though three of the four cells still "looked" proportional.
⋆ *Analogy:* independence is an *all-cells* promise, not a majority vote. It's like a contract that must hold on every clause — breaking just one clause (one cell) voids the whole agreement, no matter how compliant the rest of the document is.

Example 8 — Independence Confirmed, Continuous Case

$f_{X,Y}(x, y) = 4xy$ on $[0, 1] \times [0, 1]$. Are X, Y independent?
Solution. Marginals: $f_X(x) = \int_0^1 4xy\, dy = 2x$; $f_Y(y) = \int_0^1 4xy\, dx = 2y$ (each is a valid density on $[0, 1]$: $\int_0^1 2x\, dx = 1$ ✓). Check factorization:

$$f_X(x) f_Y(y) = (2x)(2y) = 4xy = f_{X,Y}(x, y) \quad \text{for } \textit{every } (x, y)$$

$\boxed{\text{Independent.}}$
▶ *Speed trick:* the instant you see $f(x, y) = g(x)h(y)$ *and* the support is a rectangle (no x-y coupling in the limits), you can declare independence without computing a single integral — the marginals are just g, h rescaled to integrate to 1.

Example 9 — Dependence From Support Shape Alone

$f_{X,Y}(x, y) = 2$ on the triangle $x, y > 0$, $x + y < 1$ (Example 3's density). Even though the density is *constant*, are X, Y independent?
Solution. From Example 3, $f_X(x) = 2 - 2x$, and by symmetry $f_Y(y) = 2 - 2y$. Test at $(0.2, 0.3)$:

$$f_X(0.2) f_Y(0.3) = (1.6)(1.4) = 2.24 \neq 2 = f_{X,Y}(0.2, 0.3)$$

Dependent — despite a perfectly flat, "uniform-looking" density.

⋆ *Analogy:* knowing a large order just filled on X (say x close to 1) tells you *immediately* that Y must be small, because $x + y < 1$ boxes Y into a shrinking range. It's like two people splitting a fixed-size pizza — however you slice it, the *moment* you learn one person's share, the other's possible range changes. A non-rectangular support is itself a source of dependence, no density shape needed.

Example 10 — The Rectangle Isn't Enough Either

$f_{X,Y}(x, y) = \frac{1}{4}(1 + xy)$ on $[-1, 1] \times [-1, 1]$ (a perfect rectangle). Find f_X, f_Y, and test independence.

Solution. $f_X(x) = \int_{-1}^{1} \frac{1}{4}(1 + xy)\, dy = \frac{1}{4}\,[2 + x \cdot 0] = \frac{1}{2}$ (the $\int y\, dy$ term vanishes by symmetry). Identically $f_Y(y) = \frac{1}{2}$ — *both marginals are uniform.* But:

$$f_X(x) f_Y(y) = \frac{1}{4} \neq \frac{1}{4}(1 + xy) = f_{X,Y}(x, y) \quad \text{whenever } xy \neq 0$$

Dependent — a genuine trap.

▶ *Intuition:* a rectangular support is *necessary* for independence but never *sufficient* — you must also check the density itself factors as $g(x)h(y)$ pointwise. Here both 1-D "shadows" (marginals) look innocently uniform; only the joint shape reveals the coupling, exactly the way two stocks can each look individually unremarkable while still moving together.

C. Conditional Distributions

Example 11 — Conditional PMF, Straight From a Row

Using Example 1's table, find $f_{Y|X}(y \mid X{=}1)$.

Solution. Take row $X = 1$: $(0.10, 0.15, 0.05)$, and divide by $f_X(1) = 0.30$:

$$f_{Y|X}(1 \mid 1) = \tfrac{0.10}{0.30} = \tfrac{1}{3}, \quad f_{Y|X}(2 \mid 1) = \tfrac{0.15}{0.30} = \tfrac{1}{2}$$

$$f_{Y|X}(3 \mid 1) = \tfrac{0.05}{0.30} = \tfrac{1}{6}$$

Check: $\frac{1}{3} + \frac{1}{2} + \frac{1}{6} = 1$ ✓.

⋆ *Analogy:* conditioning is *zooming into one row* of the spreadsheet and re-normalizing just that row so it becomes its own complete probability distribution — the rest of the table is simply discarded, not averaged in.

Example 12 — Conditional PDF on a Rectangle

Using Example 2's density $f(x, y) = x + y$ on the unit square, find $f_{Y|X}(y \mid x)$.

Solution. Divide the joint by the marginal $f_X(x) = x + \frac{1}{2}$:

$$f_{Y|X}(y \mid x) = \frac{x + y}{x + \frac{1}{2}}, \quad 0 < y < 1$$

Check it integrates to 1 for any fixed x: $\int_0^1 \frac{x+y}{x+1/2}\, dy = \frac{x+1/2}{x+1/2} = 1$ ✓.

▶ *Speed trick: the conditional density is always just the joint divided by the marginal, treating the conditioning variable as a fixed constant.* The resulting formula must integrate to exactly 1 over y — a free, built-in check on every conditional-density computation you ever do.

Example 13 — A Conditional Density That's Secretly Uniform

Using Example 3/9's triangle density $f(x, y) = 2$ on $x, y > 0$, $x + y < 1$, find $f_{Y|X}(y \mid x)$.

Solution.

$$f_{Y|X}(y \mid x) = \frac{2}{f_X(x)} = \frac{2}{2 - 2x} = \frac{1}{1 - x}, \quad 0 < y < 1 - x$$

This is *constant* in y over $(0, 1 - x) \Rightarrow \boxed{Y \mid X{=}x \sim \text{Uniform}(0,\ 1 - x)}$.

⋆ *Analogy:* once you know $X = x$, the triangle's constraint $y < 1 - x$ becomes a hard ceiling, but within that ceiling every value of y is equally likely — like learning how much cake is *left* after someone took a slice (x), then cutting the remainder uniformly at random. A uniform *joint* density conditions down to a uniform *conditional* density, just on a shrunken range.

Example 14 — Conditional Expectation From a Conditional Uniform

Using Example 13's result, find $\mathbb{E}[Y \mid X = x]$.

Solution. $Y \mid X{=}x \sim \text{Uniform}(0, 1 - x)$, and a Uniform$(0, M)$ has mean $M/2$:

$$\mathbb{E}[Y \mid X = x] = \boxed{\frac{1 - x}{2}}$$

At $x = 0.4$: $\mathbb{E}[Y \mid X{=}0.4] = 0.3$.

▶ *Speed trick:* once you recognize a conditional density as a *named* distribution (uniform, here), skip the integral entirely and quote its mean formula directly — this is exactly how Ch. 3.2's reservation-price and fill-probability shortcuts worked, now one level deeper (conditional on another random variable, not just a parameter).

Example 15 — A Tail Probability From a Conditional Density

Using Example 13's conditional, find $\mathbb{P}(Y > 0.5 \mid X = 0.2)$.

Solution. $Y \mid X{=}0.2 \sim$ Uniform$(0,\ 0.8)$. For a uniform, $\mathbb{P}(Y > t) = \frac{(\text{upper limit}) - t}{\text{upper limit}}$ when $0 \le t \le$ upper limit:

$$\mathbb{P}(Y > 0.5 \mid X = 0.2) = \frac{0.8 - 0.5}{0.8} = \boxed{\tfrac{3}{8} = 0.375}$$

⋆ *Analogy:* once you've identified the conditional law as Uniform$(0, 0.8)$, any probability question about it is just a *ruler measurement* — what fraction of the $[0, 0.8]$ stick lies past the 0.5 mark — no integral needed, exactly like reading a fraction off a number line.

D. Capstones — Traps & Trading Applications

Example 16 — Uncorrelated Is Not Independent

$X \in \{-1, 0, 1\}$ equally likely, and $Y = X^2$ (so $Y \in \{0, 1\}$, fully determined by X). Show $\text{Cov}(X, Y) = 0$ yet X, Y are not independent.

Solution. $\mathbb{E}[X] = \frac{-1+0+1}{3} = 0$. $\mathbb{E}[XY] = \mathbb{E}[X^3] = \frac{(-1)^3+0^3+1^3}{3} = 0$. So

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0 - 0 = \boxed{0}$$

Yet Y is a *deterministic function* of X: knowing $X = 0$ tells you $Y = 0$ for certain, while $X = \pm 1$ tells you $Y = 1$ for certain — the conditional law of Y changes completely with X, so $\boxed{\text{not independent}}$.

▶ *Full-circle intuition:* covariance only detects *linear* co-movement. Here Y swings symmetrically with $|X|$, not with X's sign, so the linear signal cancels to exactly zero while the dependence (highly nonlinear) is total. **Independent ⇒ uncorrelated always; the converse is false** — a correlation matrix showing zero is never proof that two quantities don't move together.

Example 17 — Correlated Asset Payoffs

Two assets' payoffs $X \in \{0, 10\}$, $Y \in \{0, 5, 10\}$ have joint PMF

	$Y{=}0$	$Y{=}5$	$Y{=}10$
$X{=}0$	0.30	0.10	0.05
$X{=}10$	0.05	0.20	0.30

Find $\mathbb{E}[Y]$ and $\mathbb{E}[Y \mid X{=}10]$. Does conditioning change your estimate?

Solution. Marginal: $f_Y(0) = 0.35$, $f_Y(5) = 0.30$, $f_Y(10) = 0.35 \Rightarrow \mathbb{E}[Y] = 5(0.30) + 10(0.35) = 5$. $f_X(10) = 0.05 + 0.20 + 0.30 = 0.55$. Conditional: $f_{Y|X}(\cdot \mid 10) = \left(\frac{1}{11}, \frac{4}{11}, \frac{6}{11}\right)$:

$$\mathbb{E}[Y \mid X = 10] = 5\left(\tfrac{4}{11}\right) + 10\left(\tfrac{6}{11}\right) = \boxed{\tfrac{80}{11} \approx \$7.27}$$

versus the unconditional \$5.00 — conditioning on the *other* asset's high payoff raised the estimate by ≈ 45%.

⋆ *Analogy:* this is why a trading desk never prices one asset in isolation when flows are correlated — learning Asset A paid out big is exactly like learning it's been raining all week before estimating tomorrow's river level: the joint history moves your estimate even though you haven't observed the second quantity directly.

Example 18 — The Mirror-Image Triangle

$f_{X,Y}(x, y) = c$ on $\{0 < y < x < 1\}$. Find c, both marginals, and $f_{X|Y}(x \mid y)$.

Solution. The region's area is $\frac{1}{2}$ (half the unit square), so $c = \boxed{2}$.

$$f_X(x) = \int_0^x 2\, dy = 2x, \quad 0 < x < 1$$

$$f_Y(y) = \int_y^1 2\, dx = 2(1 - y), \quad 0 < y < 1$$

$$f_{X|Y}(x \mid y) = \frac{2}{2(1-y)} = \frac{1}{1-y} \;\Rightarrow\; \boxed{X \mid Y{=}y \sim \text{Uniform}(y,\,1)}$$

▶ *Analogy:* flip Example 13's triangle across its diagonal and you get this one — same mechanics, roles of X, Y swapped. Once you know the smaller value $Y = y$, the larger value X is uniformly smeared over everything still above it, up to the ceiling at 1 — like knowing the *low* bid in an auction and asking where the *winning* bid could uniformly have landed.

Example 19 — Full Capstone: Normalize, Marginalize, Test, Condition

$f_{X,Y}(x,y) = c(x+y)$ on the rectangle $0 < x < 2$, $0 < y < 1$. (a) Find c. (b) Find both marginals and test independence. (c) Find $\mathbb{P}(Y > 0.5 \mid X = 1)$.

Solution. (a) $\int_0^2 \int_0^1 c(x+y)\,dy\,dx = c\int_0^2 \left(x+\frac{1}{2}\right)dx = c(2+1) = 3c = 1 \Rightarrow c = \boxed{\frac{1}{3}}$. (b) $f_X(x) = \frac{1}{3}\left(x+\frac{1}{2}\right)$ on $(0,2)$; $f_Y(y) = \frac{2}{3}(1+y)$ on $(0,1)$. At $(0,0)$: $f_X(0)f_Y(0) = \frac{1}{6}\cdot\frac{2}{3} = \frac{1}{9} \neq 0 = f(0,0) \Rightarrow \boxed{\text{dependent}}$. (c) $f_{Y|X}(y \mid 1) = \frac{\frac{1}{3}(1+y)}{\frac{1}{3}(1.5)} = \frac{1+y}{1.5}$. Then

$$\mathbb{P}(Y > 0.5 \mid X = 1) = \int_{0.5}^{1} \frac{1+y}{1.5}\,dy = \boxed{\tfrac{7}{12} \approx 0.583}$$

⋆ *Full-circle intuition:* every tool in this chapter, run back-to-back on one density — normalize first (find c), marginalize (sum/integrate out the other variable), test independence pointwise (one counterexample is enough to convict), then condition (divide joint by marginal) and integrate the conditional to answer a probability question. This exact four-step chain is what a Stage 1 "hard" joint-distribution question is built to test under time pressure.

Example 20 — Market-Microstructure Finale: Order Intensity & Adverse-Selection Cost

Order arrival intensity $X \in (0,1)$ and resulting adverse-selection cost $Y \in (0,1)$ have joint density $f(x,y) = c(x^2+y^2)$. (a) Find c. (b) Find $f_X(x)$. (c) Find $\mathbb{E}[Y \mid X = x]$ in closed form, and evaluate at $x = 0$ and $x = 1$.

Solution. (a) $\int_0^1 \int_0^1 c(x^2+y^2)\,dy\,dx = c\left(\frac{1}{3}+\frac{1}{3}\right) = \frac{2c}{3} = 1 \Rightarrow c = \boxed{\frac{3}{2}}$. (b) $f_X(x) = \frac{3}{2}\int_0^1 (x^2+y^2)dy = \frac{3}{2}\left(x^2+\frac{1}{3}\right) = \boxed{\frac{3}{2}x^2+\frac{1}{2}}$. (c) $\mathbb{E}[Y \mid X = x] = \dfrac{\int_0^1 y(x^2+y^2)\,dy}{x^2+\frac{1}{3}} = \dfrac{\frac{x^2}{2}+\frac{1}{4}}{x^2+\frac{1}{3}} = \boxed{\dfrac{3(2x^2+1)}{4(3x^2+1)}}$ At $x = 0$: $\mathbb{E}[Y \mid X{=}0] = \frac{3}{4}$. At $x = 1$: $\mathbb{E}[Y \mid X{=}1] = \frac{9}{16} = 0.5625$.

▶ *Full-circle intuition:* low-intensity order flow (x near 0) carries a *higher* expected adverse-selection cost here (0.75) than high-intensity flow (0.5625 at $x = 1$) — the y^2 term dominates when x^2 is small, pulling the conditional mass of Y upward. A market maker reading this relationship live would widen quotes more aggressively on *thin*, intermittent flow than on busy, continuous flow — exactly the kind of conditional-expectation reasoning Ch. 3.2 and 3.3 built, now generalized to any joint density you're handed cold.